Beyond Zero-Shot: A Systematic Quest for the Best CLIP Model for Architectural Style

A Benchmark of Adaptation Strategies from the Computer Vision Course at UNIMORE



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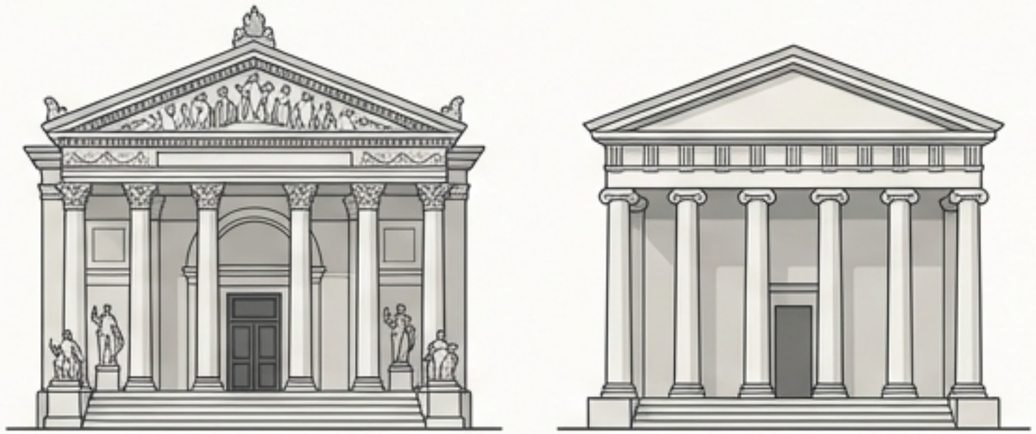
The Architect’s Eye: A Nuanced Challenge for AI

High Intra-Class Variance



All are “Gothic”

High Inter-Class Similarity



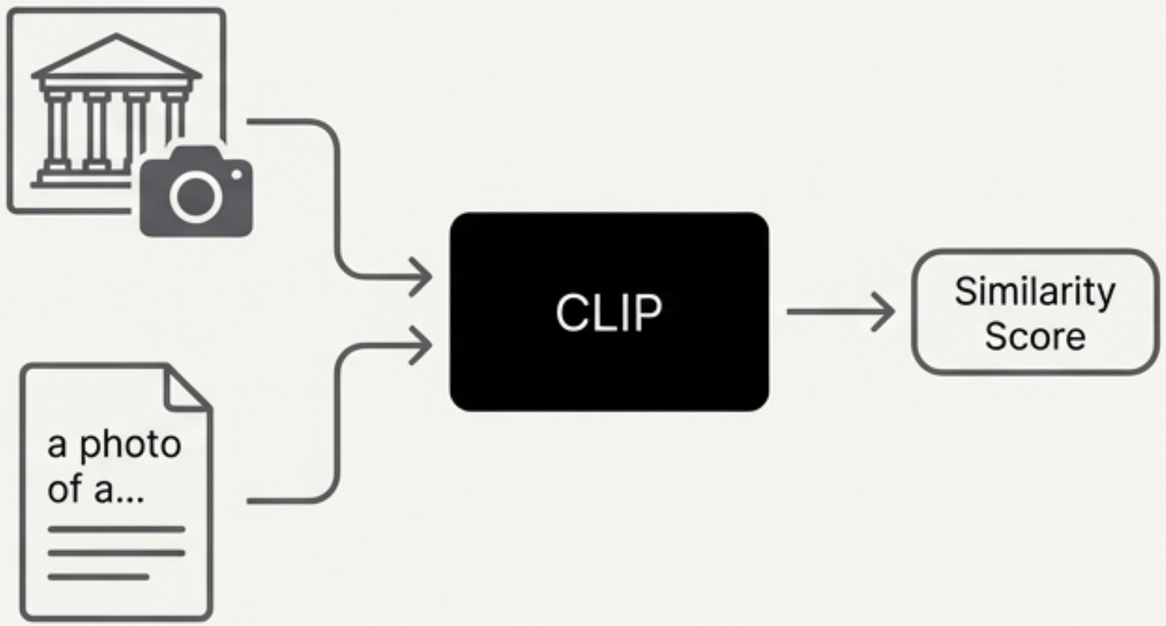
Beaux-Arts

Greek Revival

The task demands a deep understanding of style, not just object recognition. This is why we need to find the right tool.

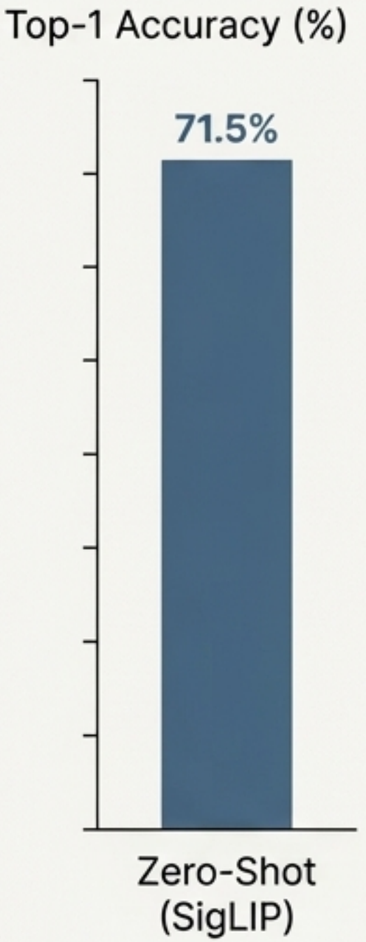
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The Starting Point: Surprisingly Strong Performance for Free



71.5% Top-1 Accuracy

Achieved by SigLIP (ViT-SO400M-14) with zero training data or fine-tuning.



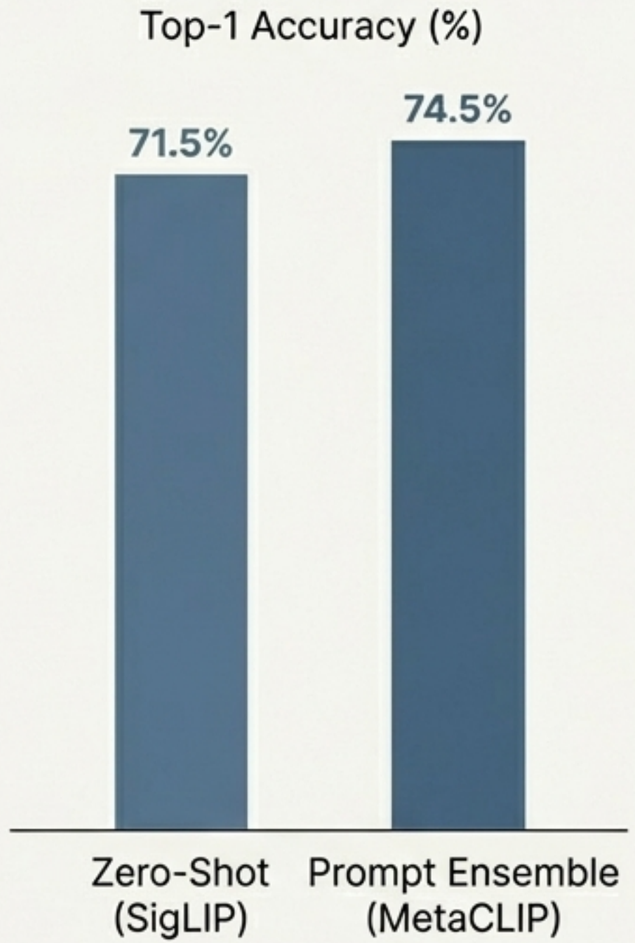
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The First Step Up: Can We Do Better With Just Words?

"a building in the {label} style"
"an example of {label} architecture"
"the facade of a {label} building"
"a photograph of a {label} structure"
"an architectural photograph of {label}"
"a style {label} building"
"the style of {label} architecture"
"a typical {label} building"
"an iconic {label} structure"

"a photo of {label} architecture"

"a picture of {label} design"
"a rendering of {label} architecture"
"a snapshot of a {label} building"
"a photo of a prchitectural {label} building"
"features of {label} architecture"
"the {label} architectural style"
"a view of a {label} building"
"a view of a {label} building"

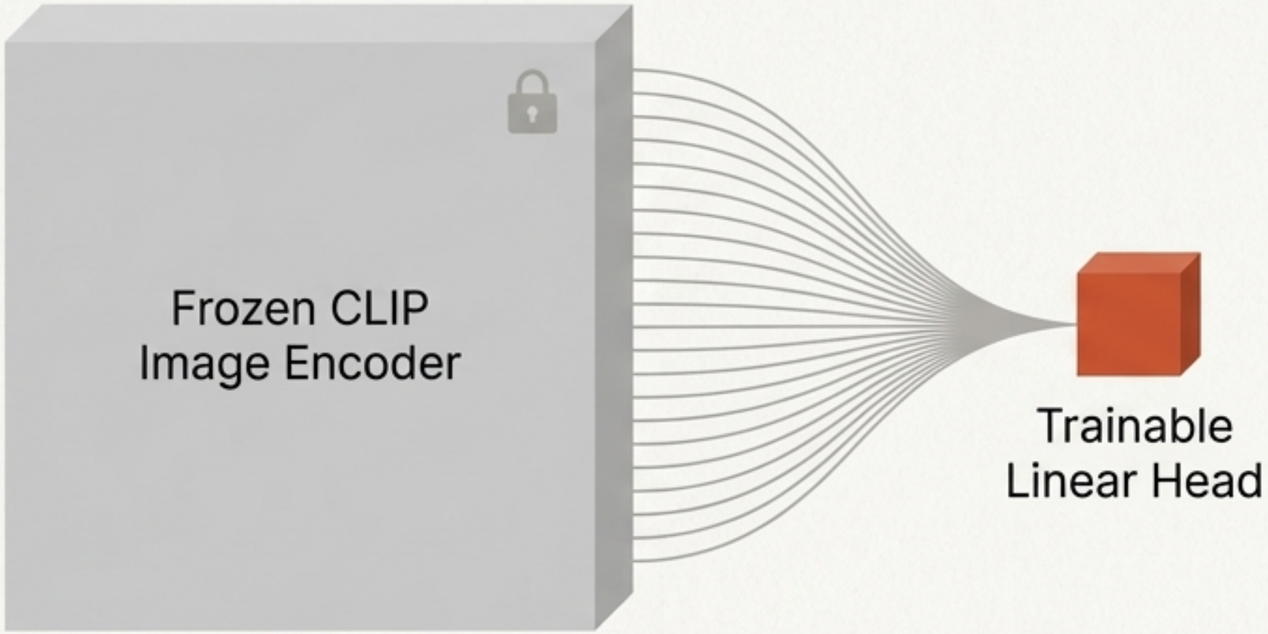


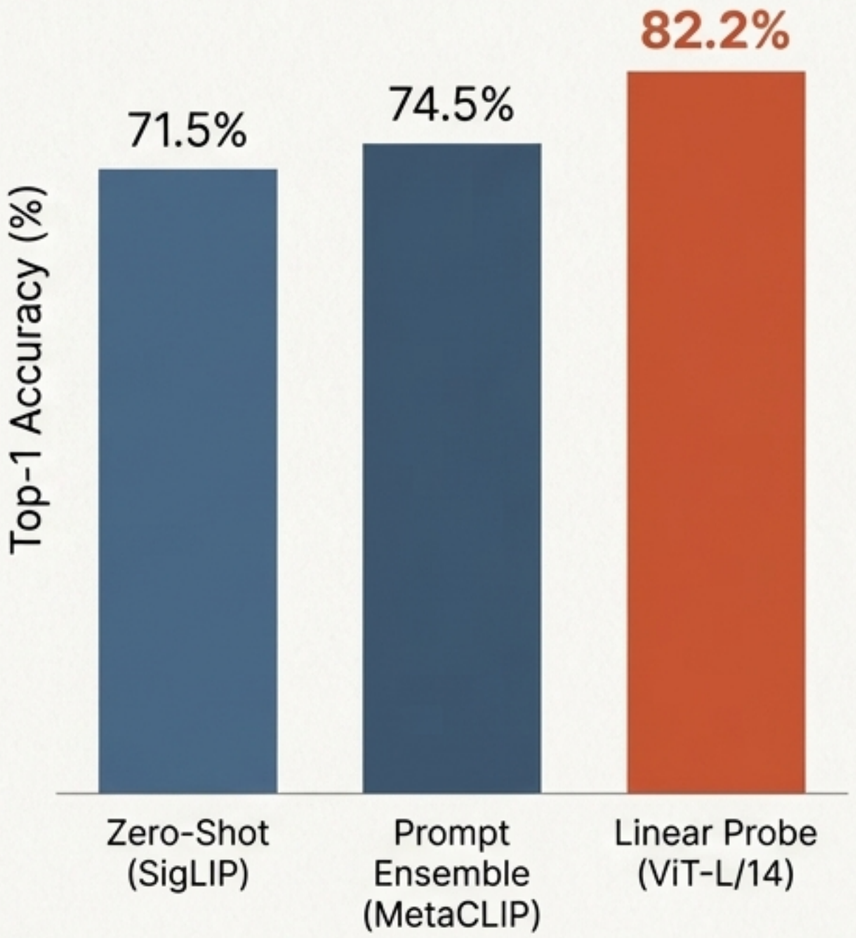
Prompt ensembling boosts our best model (MetaCLIP) by **+4.3%**, reaching **74.5% Top-1**.

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The Smart Shortcut: Learning from Frozen Knowledge

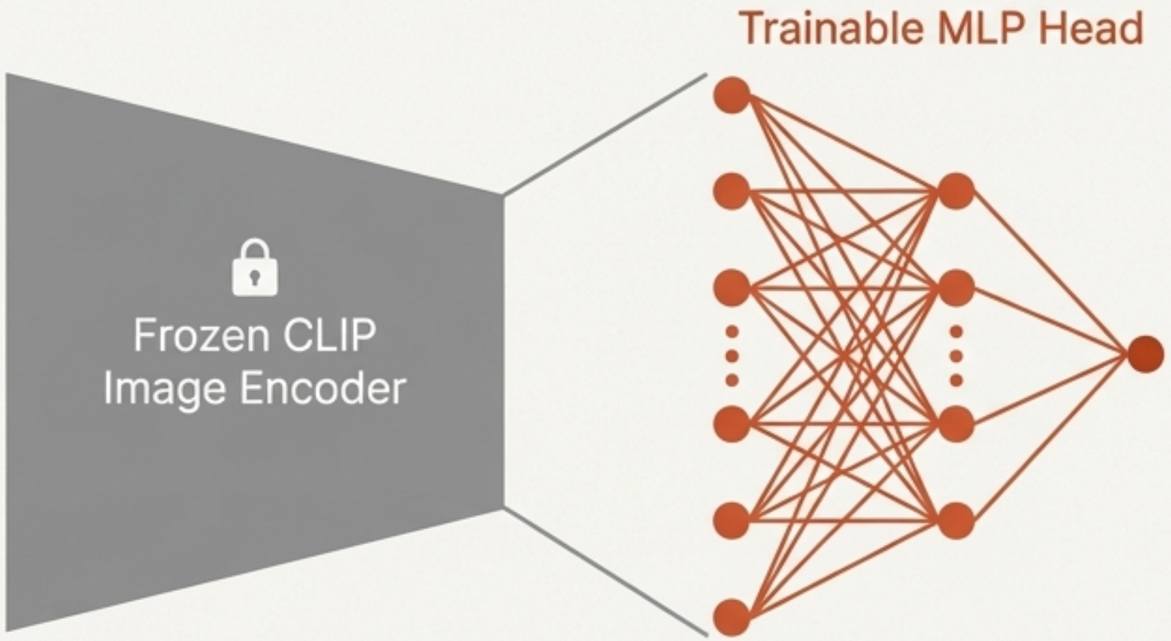


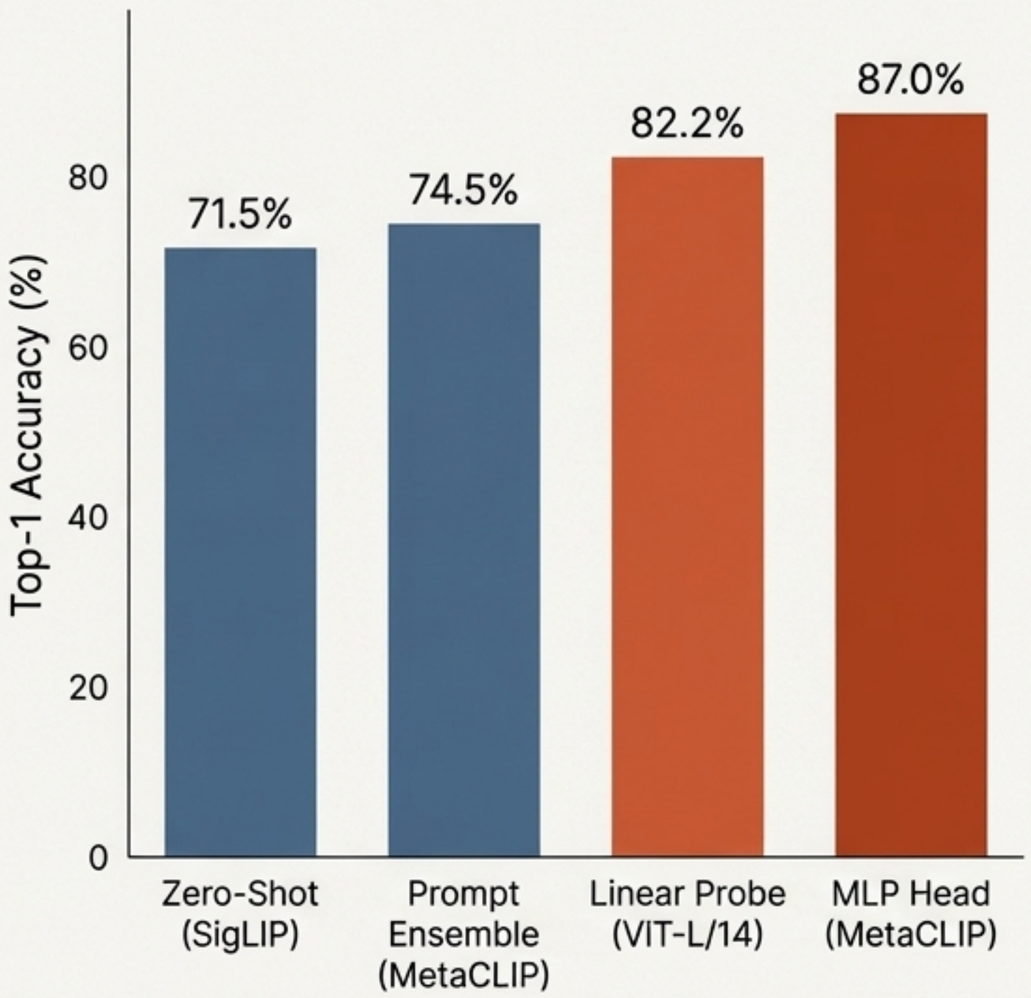


Linear Probing catapults performance. The OpenAI ViT-L/14 model jumps from **54.8% to 82.2%** Top-1.

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Squeezing Out More Power: The Non-Linear Edge





A shallow MLP adds a crucial edge, pushing our strongest backbone (MetaCLIP) to **87.0% Top-1**.

This performance is now neck-and-neck with DINOv3 (87.5%), a powerful vision-only model, validating CLIP's multimodal strength.

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Architectural Digest Editorial

Exploring a Different Path: What if the Prompt Itself Learns?



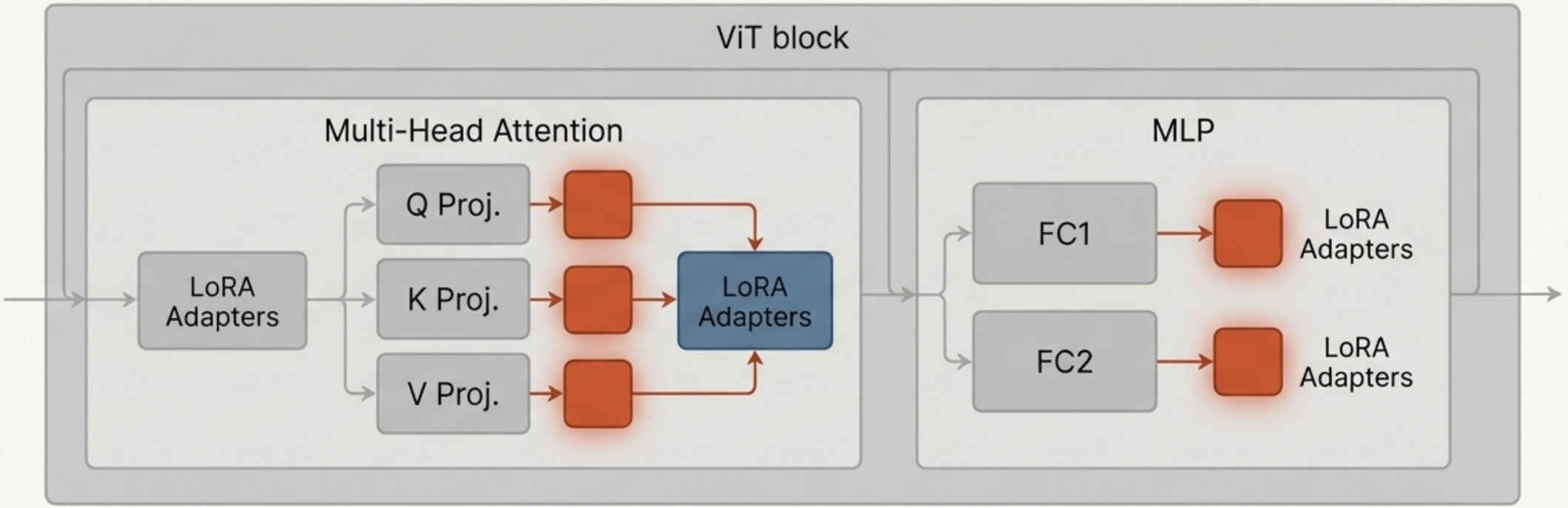
Key Result

CoOp delivers a significant boost over zero-shot (**+20.6%** for ViT-L/14), but ultimately lags behind the MLP-on-frozen-features approach for this task.

A promising direction, but for our dataset, classifying extracted features proved more effective and scalable.

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The Summit Approach: Unfreezing CLIP, Intelligently



Instead of updating all 1.8 billion weights of a model like **ViT-bigG**, we insert tiny, low-rank adapters and train ***only them***. This is parameter-efficient fine-tuning.

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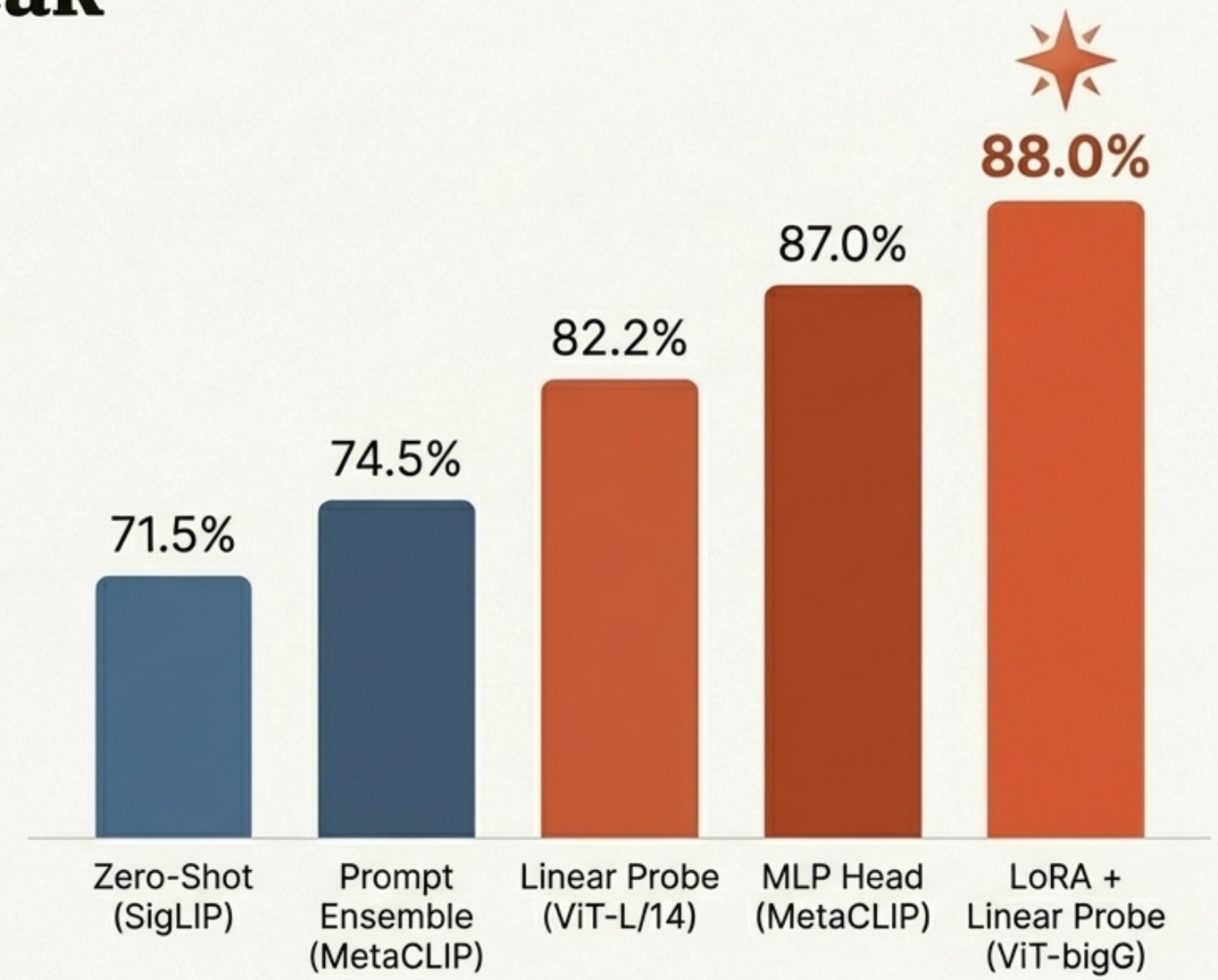
LoRA Reaches the Peak

Key Result

Fine-tuning with LoRA achieves the highest performance, pushing our strongest backbone to **88.0% Top-1 Accuracy**.

Critical Insight

The LoRA-tuned features become so linearly separable that the non-linear MLP head is no longer needed—a simple Linear Probe is sufficient.



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The Quest in Numbers: A Final Comparison

Backbone	Method	Top-1 (%)	F1-Macro (%)
ViT-bigG-14	Zero-Shot Ensemble	74.5	73.3
ViT-bigG-14	Linear Probe (Frozen)	85.9	85.9
ViT-bigG-14	MLP Head (Frozen)	87.0	87.0
ViT-bigG-14	**LoRA + Linear Probe**	**88.0**	**88.0**
DINOv3-vit7b16	Linear Probe (Frozen)	87.5	87.5

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A Practitioner's Guide to CLIP Adaptation

If you need it fast with zero training...



Use Zero-Shot with a Prompt Ensemble.

A surprisingly strong baseline with minimal effort.

If you have a little data and want the best bang-for-your-buck...



Freeze the CLIP encoder and train an MLP Head.

Stable, efficient training that unlocks massive performance gains.

If you need maximum performance and can afford light training...



Fine-tune the vision encoder with LoRA.

The state-of-the-art result, achieving the best accuracy-efficiency trade-off.

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The Key Takeaways from Our Quest

1. **Zero-shot CLIP** is a formidable baseline, but **simple prompt ensembling** provides an easy boost.

2. **Frozen features** contain immense **value**. A simple **MLP head** on a frozen backbone offers the most efficient path to high performance.

3. **LoRA** provides the ultimate performance, bridging the gap between frozen-feature extraction and full fine-tuning to surpass all other methods.

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Continue the Quest

All code, models, and experiments are publicly available to reproduce our findings and build upon them.

github.com/SalsiYari/CV_team_project

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